Modular Athletic Vest

Bao Bui, Oliver Escobar, Alexander Perez, Esei Haile
Grand Canyon University

Project Justification

- Most fitness wearables (Garmin, Fitbit, WHOOP) operate in **closed ecosystems**
- Users can only connect a **limited set of approved sensors**
- Advanced metrics (muscle activation, force, custom biomechanics) are **not supported**

Project Description

The Modular Athletic Vest is an Arduino-based wearable designed to provide **standard fitness tracking:**

- Heart rate monitoring
- Step counting
- Speed tracking
- Real-time exercise feedback based on user goals
- Workout data storage on a microSD card

Unlike traditional wearables, the vest has a **modular architecture**, allowing users to integrate **additional Arduino-compatible sensors** to measure advanced physiological or biomechanical metrics.

Target Users

- Coaches analyzing athlete performance
- Competitive athletes needing advanced training metrics
- Tech-savvy fitness enthusiasts interested in custom hardware

Design Process

The PCB layout includes connections for the Arduino R4, MPU6050 IMU, GPS module, OpenLog SD storage, and LCD interface. The design incorporates standardized mounting holes, optimized trace routing, and JST connectors for reliable communication and power distribution across the system (Fig. 1).

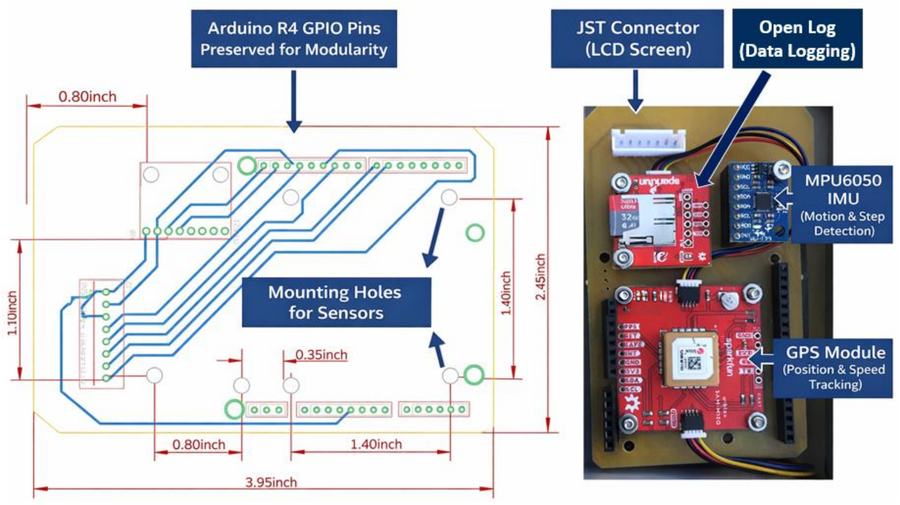

Figure 1. Custom PCB layout for an integrated sensor and microcontroller system.

System workflow from setup to output includes initialization, sensor integration, and data processing. Real-time inputs (heart rate, steps, and speed) are filtered and processed before being displayed on the LCD and stored for later analysis (Fig. 2).

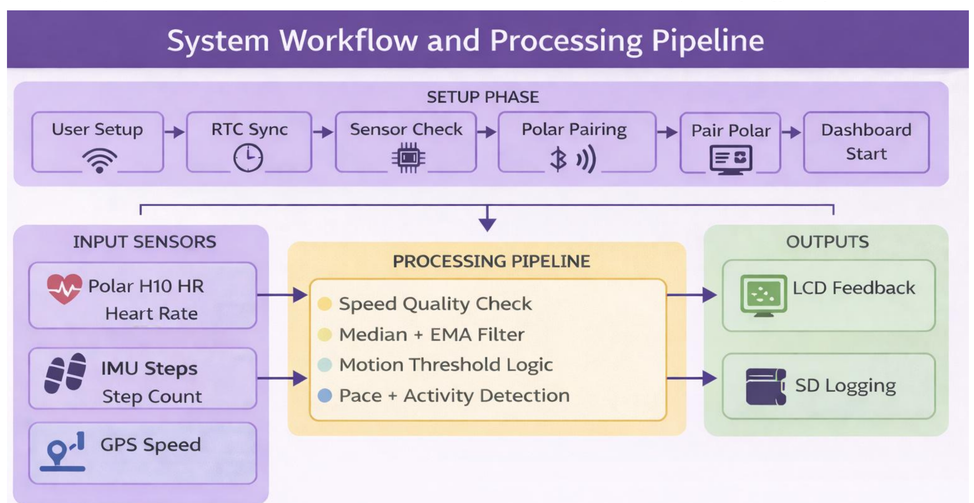

Figure 2. System workflow and real-time processing pipeline for the modular athletic vest.

A three-part enclosure houses the Arduino-PCB system and battery. The vest shows enclosure integration into the lower compartment with wiring for the LCD interface that was Velcro-mounted on a wrist-case (Fig. 3).

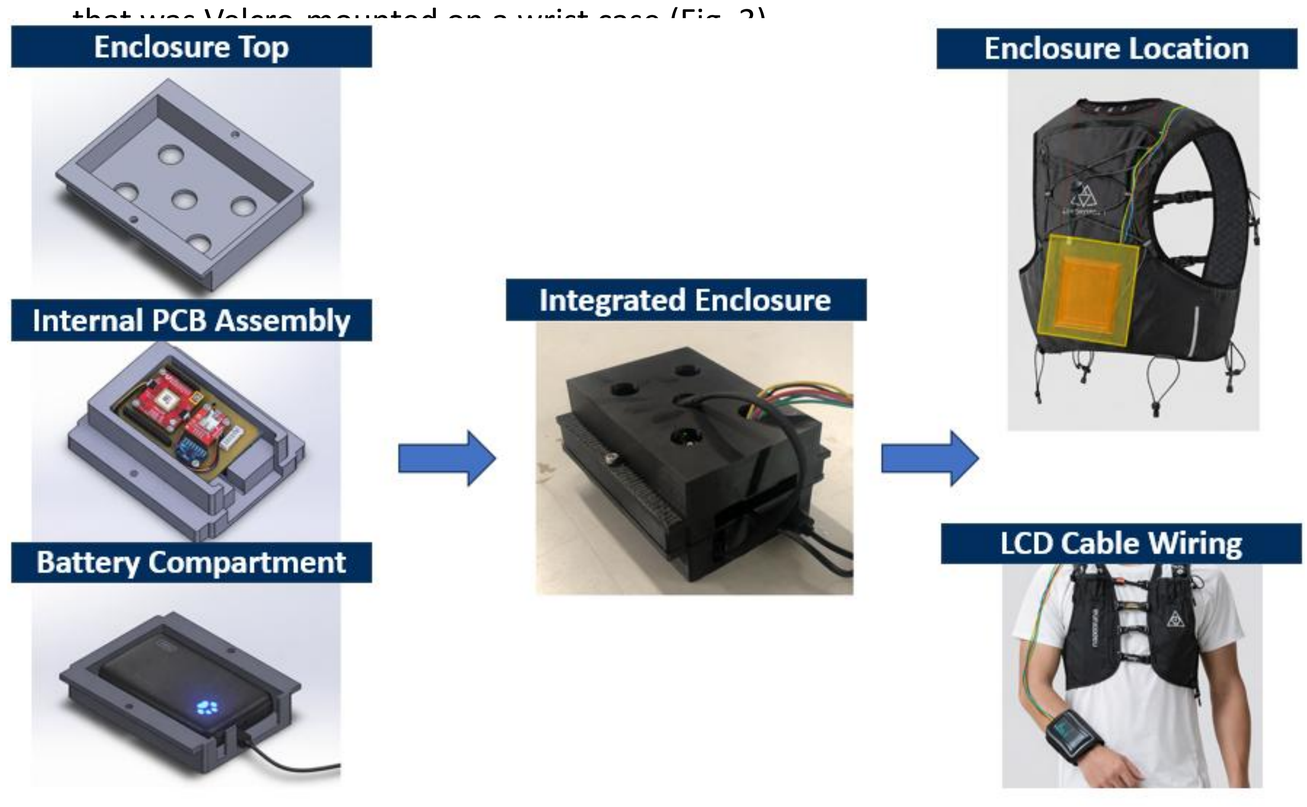

Figure 3. Enclosure design and integration into the wearable vest system.

System Performance & Validation

The functional requirements assessment confirms that the modular athletic vest meets all key performance criteria, including accuracy in heart rate (100% specificity), step tracking (MAPE = 4.69%), speed tracking (MAPE = 8.86%), battery life, data storage, and enclosure design. Validation was guided by established standards such as IEC 62133-2:2017, IEC 60601-2-47:2012, ISO 924-210:2019, and ISO 12100, with additional benchmarking methods used where formal standards were not available.

Functional Requirement	Product Specification	Engineering Standards/ Alternative Tests	Result (Pass/Fail)
Long battery life	Runtime > 24 hours	IEC 62133-2:2017	FAIL (23.5 hours)
Accurate heart rate detection	Specificity > 91%	IEC 60601-2-47:2012	PASS (100%)
Accurate step tracking	MAPE < 20% (Mora-Gonzalez et al., 2022)	Manual step count comparison	PASS (4.69%)
Accurate speed tracking	MAPE < 10% (Alphin et al., 2020)	Radar gun comparison	PASS (8.86%)
Effective user feedback system	Rating ≥ 7/10	ISO 9241-210:2019	PASS (8.69)
User comfort	Rating ≥ 7/10	ISO 9241-210:2019	PASS (7.82)
Reliable data storage	Success rate ≥ 90%	Functional storage test	PASS (100%)
Modular enclosure	≥ 2 holes ≥ 15 mm	Dimensional measurement	PASS
Secure enclosure locking	No failure under use	ISO 12100	PASS
Sweat/water resistance	Hermeticity testing	IEC 60529 ISO 105-E04	PENDING

References

- International Electrotechnical Commission. (2017). *Secondary cells and batteries containing alkaline or other non-acid electrolytes – Safety requirements for portable sealed secondary cells, and for batteries made from them, for use in portable applications – Part 2: Lithium systems (IEC 62133-2:2017).*
- International Electrotechnical Commission. (2012). *Medical electrical equipment—Part 2-47: Particular requirements for the basic safety and essential performance of ambulatory electrocardiographic systems (IEC 60601-2-47:2012).*
- International Organization for Standardization. (2019). *Ergonomics of human-system interaction—Part 210: Human-centred design for interactive systems (ISO 9241-210:2019).*
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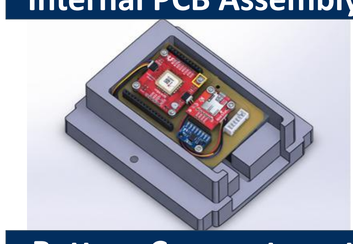
- Alphin, K. L., Sisson, O. M., Hudgins, B. L., Noonan, C. D., & Bunn, J. A. (2020). Accuracy assessment of a GPS device for maximum sprint speed. *International Journal of Exercise Science, 13*(4), 273–280.
- Mora-Gonzalez, J., Gould, Z. R., Moore, C. C., Aguiar, E. J., Ducharme, S. W., Schuna, J. M., Barreira, T. V., Staudenmayer, J., McAvoy, C. R., Boikova, M., Miller, T. A., & Tudor-Locke, C. (2022). A catalog of validity indices for step counting wearable technologies during treadmill walking: The CADENCE-adults study. *International Journal of Behavioral Nutrition and Physical Activity, 19*(1), 117. https://doi.org/10.1186/s12966-022-01350-9

Enclosure Top

Internal PCB Assembly

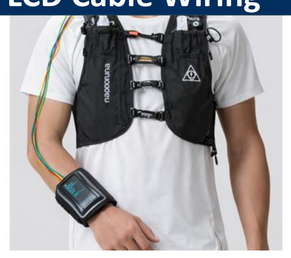


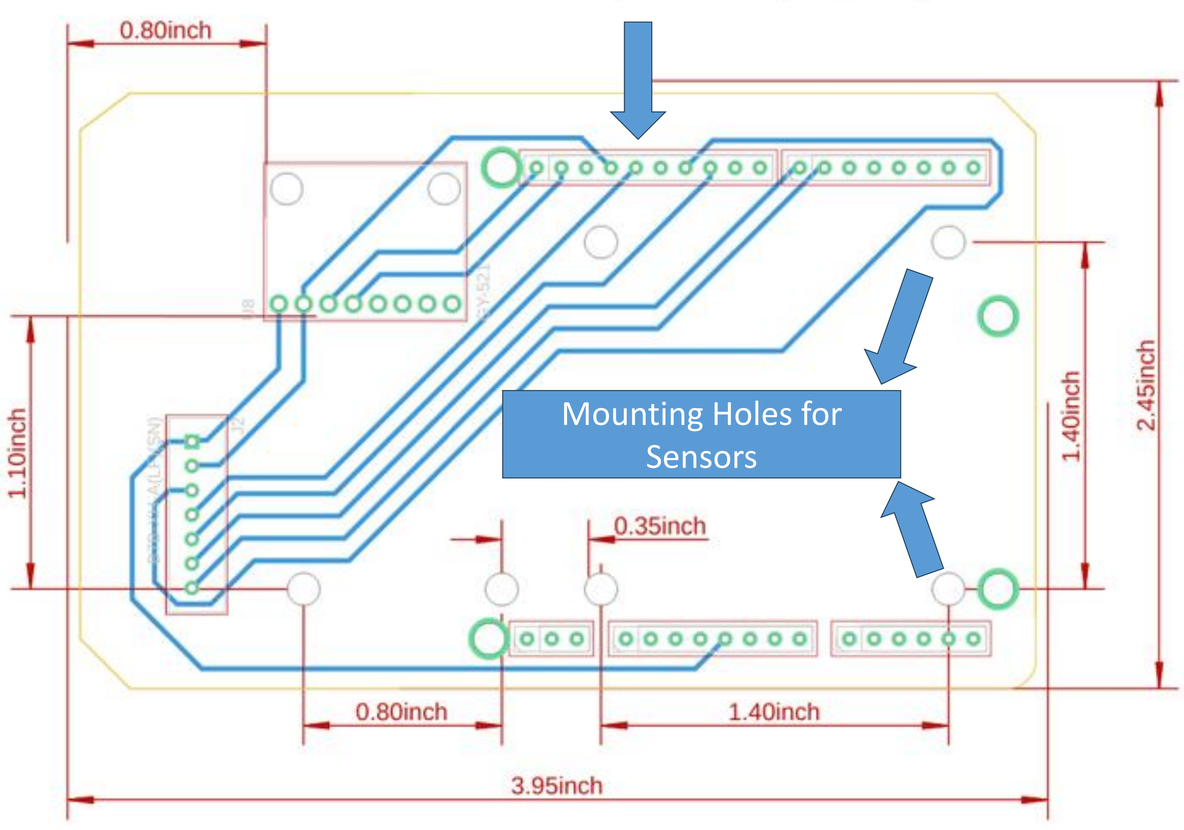
Battery Compartment

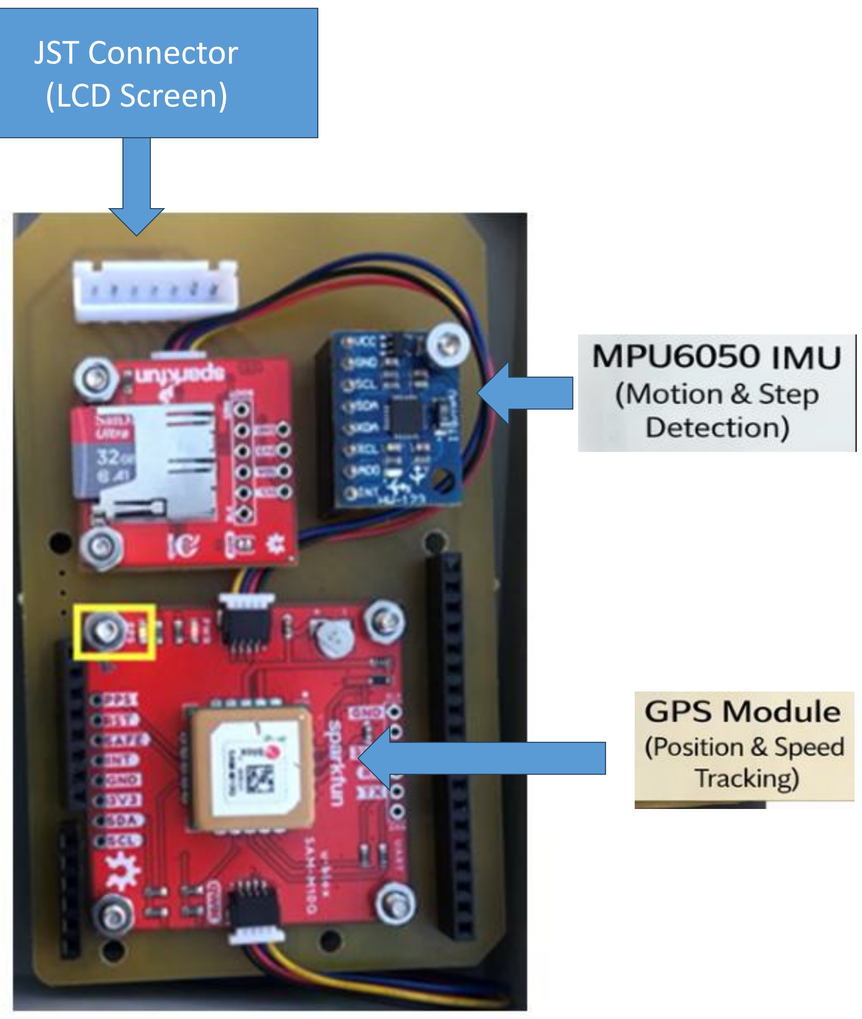
Integrated Enclosure

Enclosure Location

LCD Cable Wiring







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